

ADITYA AGARWAL

London, UK • +44 7775458618 • aditya.7@outlook.com • adityagarwal.com • LinkedIn: [aditya-5](#) • GitHub: [aditya-5](#)

PROFILE

Full-stack software engineer with 1.5+ years at Jaguar Land Rover building cloud-native IoT platforms and real-time telemetry infrastructure used by 2,000+ test rigs globally. AWS Certified Solutions Architect. Now expanding into AI/ML engineering - with hands-on experience shipping LLM-powered apps using Claude APIs and MCP tooling, and active study of RAG, QLoRA, and agent architectures. Comfortable across the full stack: edge pipelines, REST APIs, Next.js frontends, and observability dashboards.

EXPERIENCE

JAGUAR LAND ROVER

September 2024 – Present

Software Graduate Engineer

Gaydon, UK

- Built and shipped the CIARA Configuration Toolset (CCT) — a Next.js / TanStack Router platform with AWS IoT Jobs integration, enabling zero-touch software deployment to 2,000+ HiL test rigs across the UK, Hungary, China, and India.
- Designed a real-time telemetry pipeline (edge → AWS Lambda → OpenSearch → Grafana) to capture rig utilisation data, delivering dashboards that drove strategic asset decisions across global test teams.
- Resolved a critical production bug in analytics software underpinning the Jaguar Type 00 launch programme, restoring real-time reporting for 2,000+ active rigs and cross-functional stakeholder teams.
- Designed and maintained the TAMS REST API layer (GET/POST endpoints) serving as the data intake backbone for the JLR Test Platform chapter's device management infrastructure.
- Owned the CIARA Gateway — the centralised auth and routing layer built on AWS Cognito, OAuth 2.0, and Traefik, enforcing secure, unified login across all internal tooling.
- Stack: AWS (Lambda, S3, EC2, IoT Core, Greengrass), OpenSearch, InfluxDB, Grafana, Next.js, Docker, REST APIs, OAuth 2.0.

IMAGO SOFTWARE

June 2024 – August 2024

Software Developer Intern

Manchester, UK

- Delivered a cross-platform Flutter app for real-time metric visualisation from connected handheld devices, researching optimal metric calculation methods to ensure clinically meaningful UX.
- Built Flask API endpoints and optimised Docker-based AWS deployments, reducing infrastructure costs and validating the end-to-end data pipeline.

NATIONAL HEALTH SERVICE (NHS)

August 2022 – July 2023

Robotic Process Automation Engineer

Manchester, UK

- Built an Automation Anywhere bot processing up to 180,000 NHS referral letters per month, bridging two isolated clinical systems and eliminating a manual backlog — with full GDPR-compliant documentation.

OCEAN82

June 2021 – July 2022

Full-Stack Developer (Part-time)

Bolton, UK

- Architected a MERN stack stock-trading simulation platform — 4 microservices, real financial models, and AWS infrastructure (Route53, load balancing) — supporting live 2-hour trading sessions.
- Scaled a judging platform for the World Communication Awards 2021 to 5× capacity by replacing singleton DB connections with connection pooling; zero downtime during the live event.

UNIVERSITY OF MANCHESTER

July 2022 & July 2023

Software Engineering Intern

Manchester, UK

- Trained an ML fact-verification model on 20 GB of scraped data, deployed as a React app; built a Flutter ML recipe app presented at a data science conference.

EDUCATION

University of Manchester, UK

July 2024

BSc. (Hons) Computer Science with Industrial Experience | First Class, 75%

Delhi Public School, Kanpur, India

May 2019

A-Level Equivalent (AISCSE): 96.5% | GCSE Equivalent (AISCSE): 10/10 CGPA

CERTIFICATIONS

AWS Certified Solutions Architect Associate (SAA-C03)

August 2025

TECHNICAL SKILLS

AI / LLM Tooling: Anthropic Claude API, Claude Code, RAG patterns, context engineering

Cloud & Infra: AWS (Lambda, S3, EC2, IoT Core, Greengrass), Docker, CI/CD

Backend: Node.js, Python, FastAPI, Flask, REST APIs, OpenSearch, InfluxDB, MongoDB, PostgreSQL, MySQL

Frontend: Next.js, React, TypeScript, JavaScript, Flutter, Angular

Observability: Grafana, Tableau, Elastic Stack

Auth & Security: AWS Cognito, OAuth 2.0, Traefik, Auth0

SELECTED PROJECTS

WHATSAPP CRM & BROADCAST DASHBOARD *React, FastAPI, PostgreSQL, Meta Graph API, WebSockets,*

- Built a full-stack real-time CRM dashboard integrating the WhatsApp Cloud API, enabling two-way customer conversations, bulk broadcast campaigns with pre-approved templates, and live delivery analytics.
- Architected a high-performance FastAPI backend with fully async endpoints and background tasks to handle high-volume Meta webhook payloads without blocking the main event loop.
- Established a WebSocket layer between React and FastAPI for instantly pushed bi-directional event updates (Sent → Delivered → Read); deployed on Vercel (frontend) + Koyeb (containerised backend) + Neon PostgreSQL.
- Utilised Antigravity throughout development, using agentic workflows for schema design, async migration debugging, and API integration.

OCTAVE PLATFORM *React, Bloomberg API, TypeScript*

- Built frontend components for a fintech equity research tool; integrated live Bloomberg Terminal data and ESG sentiment visualisations for ethical investment teams.

ASAAAYE.COM *React → Shopify migration, AWS (EC2, S3, CloudFront, Aurora)*

- Designed and deployed a full-scale luxury fashion e-commerce platform with auto-scaling, low-latency global asset delivery, and responsive design across mobile and web.

BERT-CRF FOR NAMED ENTITY RECOGNITION *Python, PyTorch, HuggingFace*

- Final-year dissertation: implemented and evaluated a BERT-CRF model against standard BERT across three NER domains (CoNLL2003, FINER, BC5CDR). Outperformed baseline on general and financial NER; findings highlighted the role of domain specificity in NLP model performance. Codebase built for full reproducibility and cross-dataset extensibility.

AWARDS & ACTIVITIES

- Netcraft Prize 2021 — Top 10 of 400 Year 1 Computer Science students, University of Manchester.
- Digital Media Lead, IWC Schaffhausen (9ByIWC) — managed brand web presence and product photography for a luxury Swiss watch label (2021–2023).
- Scholar Badge Holder for 11 consecutive years; awarded the Blue Coat & Blue Tie for academic excellence.